ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q3, Q4

Question 1

(a) (i) $f_x = f/p = 16\text{ mm} / 5\ \mu\text{m} =$ **3200 px**.

(ii) Sensor width = $1280 \times 5\ \mu\text{m} = 6.4$ mm; FOV = $W_s \cdot Z/f = 6.4 \times 400/16 =$ **160 mm**.

(b) $\text{LSB} = V_{range}/2^n = 5\text{ V} / 2^{12} = 5/4096 = 1.2207$ mV.

Smallest resolvable force $\Delta F = \text{LSB} / S = 1.2207\text{ mV} / (10\text{ mV/N}) =$ **0.1221 N**.

(c) Using forward Euler: $x_{k+1} = x_k + T(A\,x_k + B\,u_k)$, so $F = 1 + AT$ and $G = BT$.

$F = 1 + (-2)(0.1) =$ **0.8**, $G = 1 \times 0.1 =$ **0.1**.

$x_{k+1} = 0.8\,x_k + 0.1\,u_k$.

Question 3

(a) (1) Differentiation amplifies noise in the encoder signal.

(2) Finite differencing needs a lot of memory and processing, wich is slow on a micro-controller at 1 kHz.

(b) $\omega_c = 2\pi \times 20 = 125.66$ rad/s; Tustin: $s = (2/T)(z-1)/(z+1)$; $a = \omega_c T/2 = 0.0628$.

$y_k =$ **$0.0591\,(u_k + u_{k-1}) + 0.8818\,y_{k-1}$**.

DC gain is unity becuase it is a low-pass fliter.

(c) $A - LC = [-\ell_1, 1; -\ell_2, -2]$; characterstic polynomial $\det(sI - A + LC) = (s + \ell_1)(s+2) + \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 + \ell_2)$.

Desired $(s+10)^2 = s^2 + 20s + 100$. Then $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$, and $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 100 - 18 = 82$.

$L = [18, 82]^T$.

(d) Advantage of the obsever: it uses the model and the input u, so there is no lag in the estimate, which a filter always introduces.

The filter is safer when the sampling rate is very high, becuase the observer would not be able to run fast enough.

Question 4

(a) $\ddot{\theta} + a\dot{\theta} + c\theta = b(N_{ff}\,r - k_1\theta - k_2\dot{\theta}) \Rightarrow \ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\,r$.

With $a = 2$, $b = 1$, $c = 6$, $k_1 = k_2 = 2$: $s^2 + 4s + 8 = 0 \Rightarrow s = -2 \pm 2j$ (stable).

DC gain $= bN_{ff}/(c + bk_1) = N_{ff}/8 = 1 \Rightarrow$ **$N_{ff} = 8$**.

(b) $G_{DC} = bN_{ff}/(c + bk_1)$. Numerator: $0.75 \times 8 = 6$. Denominator: $c + k_1 = 8$ as before.

$\theta_{ss} = 6/8 =$ **0.75**, so error = **25%**.

(c) Substituting the control law and differentiating once: $s^3 + (c + bk_2)s^2 + (a + bk_1)s + bK_I$.

Desiered: $(s+4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

$6 + k_2 = 8 \Rightarrow$ **$k_2 = 2$**; $2 + k_1 = 24 \Rightarrow$ **$k_1 = 22$**; $K_I =$ **32**.

(d) For a constant r, in steady state all derivatives vanish: $\dot{\xi} = r - \theta_{ss} = 0 \Rightarrow \theta_{ss} = r$, independent of b and c; the closed loop only needs to be stable so that a steady state exists.

Failure mode: integrator windup. When the actuator saturtes the error persists, ξ keeps growing and causes a large overshoot afterwards. Remidy: anti-windup (clamp the integrater, conditional integration or back-calculation).